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REVIEW

Chronic Traumatic Encephalopathy Neuropathologic Change Is Not Associated With Suicide in Former Athletes

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ABSTRACT

There are important knowledge gaps and misunderstandings relating to chronic traumatic encephalopathy (CTE). A systematic review and meta-analysis, published in April of 2025, was designed to examine the prevalence, risk factors, and potential outcomes of CTE neuropathologic change (CTE-NC) in former athletes and concluded that suicide was a potential outcome of CTE-NC (Qi et al., *Scandinavian Journal of Medicine & Science in Sports*. 2025;35(4):e70047). The review included eight eligible postmortem studies, four of which were used to estimate a possible association between the postmortem neuropathology and suicide as a manner of death. The authors of two of those four studies did not report a statistically significant association between suicide as the manner of death and having CTE-NC identified in brain tissue. In the largest of the four studies, the brain donors with CTE-NC were statistically significantly less likely to have suicide as their manner of death than donors who did not have CTE-NC. Considering the four studies included in the systematic review, other studies of CTE-NC not included in the review, and the broader literature relating to suicidality in former athletes, the best available evidence suggests that CTE-NC is not associated with depression, suicidality, or suicide.

1 | Introduction

The Concussion in Sport Group held an international consensus conference in October of 2022 in Amsterdam [1]. The conference focused on many topics, such as the assessment, medical management, and rehabilitation of sport-related concussion. The conference also addressed possible long-term mental health problems, neurological conditions, and neurodegenerative diseases in former athletes [2]. A consensus statement arising from the work leading up to and during the conference was published in 2023 [3]. Important knowledge gaps relating to chronic traumatic encephalopathy neuropathologic change (CTE-NC) and traumatic encephalopathy syndrome (TES) [2, 3] were identified. A quote from the consensus statement regarding these knowledge gaps is reprinted below.

> The prevalence of CTE-NC (a neuropathological entity) and TES (a clinical diagnosis) in former athletes, military veterans and people from the general population is not known. It is also not known

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> whether (1) CTE-NC causes specific neurological or psychiatric problems, (2) the extent to which CTE-NC can be clearly identified within the presence of Alzheimer's disease neuropathology or (3) whether CTE-NC is inevitably progressive (p. 705) [3].

At the end of the international conference, the audience was invited to share their views on priorities for future research. There were 342 participants who prioritized their top five topics for future research, with possible long-term effects on brain health ranking first [3].

2 | Systematic Review and Meta-Analysis of CTE-NC and Suicide

A recently published systematic review and meta-analysis [4] attempted to address knowledge gaps relating to CTE-NC. The authors identified eight eligible postmortem studies, published between 2015 and 2024, and concluded that the 'pooled prevalence' of CTE-NC was 53.7% [4]. The authors reported that across the eight post-mortem studies that they reviewed, the percentages of the samples with CTE-NC ranged from 26.9% to 91.7%. Half of the studies had sample sizes of fewer than 50 subjects, and only two studies included more than 100 subjects. To estimate prevalence in a population, the sample ideally should be randomly selected from the population and be representative of the population. The samples included in the systematic review cannot be considered to reflect the population of former athletes—they are highly selected and biased samples. The authors acknowledge this limitation in their Discussion and reported that sample selection bias could have resulted in an overestimation of the prevalence of CTE-NC [4]. The authors also acknowledged the limitations of some small sample sizes, heterogeneity among the samples, and the possibility of publication bias.

The authors did not include several larger studies that reported much lower proportions of the samples with identified CTE-NC. In 2020, Bieniek and colleagues screened 750 brain donation cases, 300 former athletes and 450 who were not athletes, and reported that CTE-NC was present in only 15 (5.0%) of the former athletes and 6 (1.3%) of those who were not athletes [5]. Bieniek also coded cases that resembled CTE-NC, but did not meet criteria for CTE-NC, and indicated that they had 'features' of CTE-NC. Combining cases with identified CTE-NC and features of CTE-NC, then 27 of the former athletes (9.0%) and 15 of those who were not athletes (3.3%) had neuropathology resembling CTE-NC. Sport-specific proportions for those participating in men's contact, collision, and combat sports were provided in the online supplementary material. Those with CTE-NC were as follows: football = 7.9% (11/140), hockey = 5.0% (1/20), soccer = 0% (0/5), wrestling = 2.6% (1/39), and boxing = 28.6% (2/7); compared to those who did not participate in sports = 2.4% (6/245). Considering men only, from those specific sports (football, hockey, soccer, wrestling, and boxing), 7.1% (15/211) had CTE-NC. There were also 10 women who participated in hockey, football, or soccer, and none had identified CTE-NC.

In 2022, Priemer and colleagues examined brain tissue donated by the families of 225 military service members and veterans and identified CTE-NC in 10 cases (4.4%) [6]. This study might not have been included in the systematic review because it focused on service members and veterans. There were 60 service members with a history of participating in contact, collision, and combat sports, and 10 had CTE-NC (16.7%). In 2023, Iverson and colleagues examined for CTE-NC in brain tissue donated from 186 men and no cases had definitive CTE-NC. In that sample, there were 58 men who had a personal history of participating in contact or collision sports. For comparison, Qi and colleagues reported that the percentages with CTE-NC from the eight studies that they reviewed ranged from 26.9% to 91.7%, whereas the percentages from the three studies above, that they did not include, ranged from 0% to 16.7%. Combining the men who participated in contact, collision, or combat sports from the three studies above, 7.6% (25/329) had CTE-NC, whereas Qi and colleagues reported that the 'pooled prevalence' was 53.7% [4].

2.1 | The Authors Concluded That Suicide Is a Potential Outcome of CTE-NC

The authors concluded that suicide is a 'potential outcome' of having CTE-NC (see the Abstract of the systematic review) [4]. In their conclusions section, the authors wrote: 'Furthermore, the identification of suicide as a potential outcome of CTE underscores the severe mental health implications associated with this disease' (p. 11) [4]. These conclusions were based on their analyses of four studies.

The authors of the systematic review included a case series from Canada involving 52 former professional athletes, with a total of 19 cases in the series having CTE-NC (36.5%) of whom 14 (26.9%) had low burden CTE-NC [7]. In that study, four of 11 cases (36.4%) who did not have CTE-NC or a neurodegenerative disease had suicide as their manner of death. Of those with CTE-NC and no neurodegenerative disease, seven of 14 (50%) had suicide as their manner of death (see Table 1 in the article by Taskina and colleagues) [7]. The authors of the systematic review also included a small case series from Australia [9]. A study by Suter and colleagues identified CTE-NC in 12 of 21 brain donors (57.1%) [9]. Suter and colleagues noted that six of 12 (50%) donors with CTE-NC had suicide as their manner of death, compared to one of nine donors (11.1%) who did not have CTE-NC—leading them to speculate that CTE-NC might be a 'risk factor' for suicide [9].

The largest of the four studies included in the systematic review was by Daneshvar and colleagues; they studied 631 brain donors with a history of playing football [11]. In Table 1 of that article, suicide as a manner of death was presented. Those findings are illustrated in Figure 1. In that study, 26.3% (46/180) who did not have CTE-NC had suicide as their manner of death compared to 13.7% (62/451) who had CTE-NC. Therefore, the brain donors with CTE-NC were statistically significantly *less likely* to have suicide as their manner of death than donors who did not have CTE-NC [$\chi^2(1, 631) = 12.65$, $p < 0.001$, odds ratio (OR) = 0.64, 95% confidence interval (CI) = 0.30–0.71].

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TABLE 1 | Comparing those with and without chronic traumatic encephalopathy neuropathologic change on suicide as a manner of death.

First author	Year	Brain donation program	CTE-NC		No CTE-NC	
			N	Suicide (%)	*N*	Suicide (%)
McKee [8][b]	2013	Center for the Study of Traumatic Encephalopathy Brain Bank	7/64	10.9	6/18	33.3
Bieniek [5]	2020	Mayo Clinic Tissue Registry	1/41	2.4	34/436	7.8
Suter [9][a]	2022	Australian Sports Brain Bank	6/12	50.0	1/9	11.1
Iverson [10]	2023	Lieber Institute for Brain Development	4/10	40.0	63/176	35.8
Taskina [7][a]	2023	The Hospital for Sick Children and University of Toronto	7/14	50.0	4/11	36.4
Daneshvar [11][a,b]	2023	Understanding Neurologic Injury and Traumatic Encephalopathy and Framingham Heart Study Brain Banks	62/451	13.7	46/180	26.3
McKee [12][a,b]	2023	Understanding Neurologic Injury and Traumatic Encephalopathy Brain Bank	33/63	52.4	54/89	60.7
Abdolmohammadi [13][b]	2024	Understanding Neurologic Injury and Traumatic Encephalopathy and Framingham Heart Study Brain Banks	12/42	28.6	10/35	28.6

Abbreviations: CTE-NC, chronic traumatic encephalopathy neuropathologic change; *N*, sample size.
[a]Studies included in the systematic review by Qi and colleagues [4].
[b]Some of the same subjects might be included in more than one of these studies.

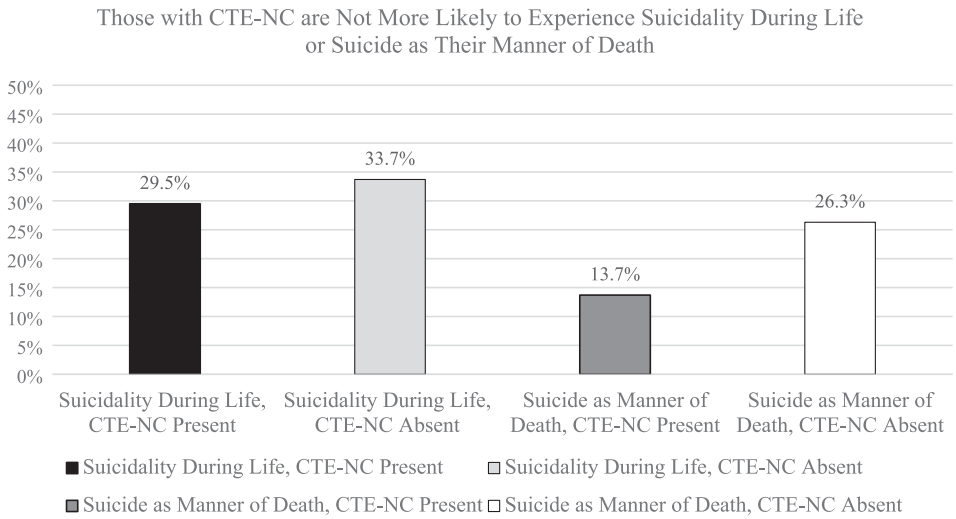

FIGURE 1 | Comparing those with and without chronic traumatic encephalopathy neuropathologic change on suicidality during life and suicide as a manner of death. The data for suicidality during life are from Mez et al. [14] and the data for suicide as the manner of death are from Daneshvar et al. [11] For suicidality, data were derived from 336 consecutive brain donors exposed to repetitive head impacts from sports, military service, and/or physical violence, 244 (72.6%) of whom were identified as having CTE-NC and 92 (27.4%) of whom did not have CTE-NC. See eTable 5 of the online supplement for Mez and colleagues. The data for suicide as a manner of death are from Daneshvar and colleagues [11]. They studied 631 brain donors with a history of playing football. There were 451 (71.5%) who were identified as having CTE-NC and 180 (28.5%) who did not have CTE-NC.

The second largest study in the systematic review was by McKee and colleagues [12]. McKee and colleagues published a postmortem study of 152 brain donors who died prior to the age of 30 [12]. The authors reported that 63 donors, 41% of the sample, had postmortem evidence of CTE-NC—with 95% (60/63) having 'mild' stage CTE-NC. More than half had suicide as their manner of death. There was no difference between those with (33/63; 52.4%) and without CTE-NC (54/89; 60.7%) in suicide as their manner of death.

Of the four studies included in the systematic review [4], the largest study [11] showed that those with CTE-NC were less likely, not more likely, to have suicide as their manner of death. The second largest study [12] showed no association between

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suicide as a manner of death and having CTE-NC. Some studies that have addressed the association between CTE-NC and suicide, that were not included in the systematic review, are summarized below.

2.2 | Neuropathology Studies Not Included in the Systematic Review

In 2013, McKee and colleagues published an important study relating to the neuropathology of CTE. This study was not eligible for inclusion in the systematic review because it was published before 2015. In that study, 6/18 (33.3%) athletes or former athletes with no CTE-NC completed suicide compared to 7/64 (10.9%) athletes or former athletes with CTE-NC [$\chi^2(1,82) = 5.28$, $p = 0.022$, OR = 4.07, 95% CI = 1.16–14.29]. In that autopsy case series, those who did not have CTE-NC were statistically significantly *more* likely to have completed suicide than those who had CTE-NC.

In 2020, Bieniek and colleagues published a large postmortem study of former amateur athletes, compared to those who did not participate in sports, and identified CTE-NC in 2.8% of the total sample (21/750) [5]. They also identified another 21 cases (2.8%) who had 'features of CTE-NC.' Thus, there were 42 cases that had CTE-NC or features of CTE-NC (5.6%), 41 of whom were men. In that study, the percentages of men with either CTE-NC or features of CTE-NC ($n = 41$) were statistically compared to the large sample of men who did not have CTE-NC ($n = 436$). There was no significant difference between those two groups in suicide as their manner of death [1/41 (2.4%) with CTE-NC or features of the pathology compared to 34/436 (7.8%) with no CTE-NC having suicide as their manner of death].

In 2021, researchers from Boston University published a large-scale clinicopathological study [14] that was designed to validate the 2014 diagnostic criteria for traumatic encephalopathy syndrome (TES) [15]. Suicidality was part of the 2014 diagnostic criteria for TES. The study included data from 336 brain donors who were exposed to repetitive head impacts from sports, during military service, and/or from physical violence. There were 244 (72.6%) who were identified as having CTE-NC and 92 (27.4%) who did not have CTE-NC. There was no association between having CTE-NC and suicidality. Specifically, suicidality during life was present in 29.5% with CTE-NC and 33.7% in those who did not have CTE-NC (see eTable 5 of the online supplementary material for Mez et al. [14]). These findings are illustrated in Figure 1. This study might not have been included in the systematic review because the authors wanted to avoid analyzing data from more than one published study examining data from the same sample, and they included a larger study in the systematic review from researchers from Boston University involving 631 brain donors with a history of playing football [11].

In 2023, Iverson and colleagues published a postmortem studying involving 186 male brain donors, 25.8% of whom had a personal history of playing amateur football and 36.0% who had suicide as their manner of death [10]. There were no cases of definitive CTE-NC. There were 10 cases, 5.4% of the sample, who were rated as having 'features' of CTE-NC (as defined by Bieniek et al. [5]). Of the 10 cases with features of CTE-NC, four (40%) had suicide as their manner of death. Of the 176 cases that did not have CTE-NC, 63 (35.8%) had suicide as their manner of death.

A study published in 2024 examined a sample of 77 brain donors who had a personal history of participation in ice hockey [13]. This study was not eligible for inclusion in the systematic review because it was published after the cutoff date for the literature search. Approximately half of the former hockey players (42/77, 54.5%) had CTE-NC. Regarding manner of death, there was no difference in suicide (29% in those who had CTE-NC and 29% in those who did not have CTE-NC [13]).

3 | Cohort Studies Relating to Suicide in Former Athletes

All the studies reviewed above, and the studies included in the systematic review by Qi and colleagues [4], were case series or cross-sectional studies. Case series and cross-sectional studies should not be used to infer causal associations [16], but they can generate hypotheses to be investigated in more rigorous longitudinal study designs, such as cohort studies [17, 18]. Iverson and colleagues conducted a systematic review of cohort studies relating to later-in-life problems with brain health in former athletes [2]. They identified nine cohort studies that included data allowing for the examination of either suicidality during life or suicide as a manner of death [19–27], and none of the studies revealed an association between playing contact or collision sports and future risk for suicidality or suicide. In one of those studies, Lehman and colleagues [25] examined suicide mortality in former National Football League players. Of the cohort of 3493 former players, 537 had died. Of the 537, 12 had suicide as their manner of death (2.2%). The authors reported that the mortality rate for suicide in former NFL players was approximately half of that expected from men in the general population. Considering five cohort studies that examined former professional athletes (i.e., football or soccer), all reported either significantly lower risk for suicide [23–25] or no difference compared to men in the general population [26, 27].

4 | Suicide has Tragic Consequences

Suicide is a tragedy for families, friends, teams, communities, and society—the impact can be felt generationally—and the consequences extend well beyond those most deeply and immediately affected. A large percentage of adults in the general population have lifetime exposure to suicide [28–30]. One study estimated that for each person who completed suicide, there was an average of 135 people exposed (i.e., who knew the person) [31]. Adults who are exposed to suicide are more likely to experience depression [28], anxiety [28] and suicidal ideation [28, 32, 33]. Bereavement following someone's suicide is associated with increased risk for attempting suicide in young adults [34]. Being closer to the decedent as a family member, but also perceived psychological closeness, is associated with, as expected, greater adverse psychological effects [28, 35–37]. Risk

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for suicidal ideation in those exposed to suicide might be greater in men than women [35, 36].

People who experience sudden bereavement following the sudden death of a friend or relative by suicide can feel highly stigmatized by this loss [38], and this stigmatization is associated with greater suicidality [39]. A small subgroup of people who have experienced the death of a close contact by suicide might be prone to view suicide as a more tangible option for themselves (whether it is feared or not); they might identify with and think about the deceased often, and they might have heightened perceptions and awareness of shared vulnerabilities to suicide [40].

Similarly, for active duty military service members and veterans, a large percentage have exposure to suicide [41–44], and those exposed to suicide are more likely to experience depression [41, 42], anxiety [41], traumatic stress [42], and suicide ideation [41, 42, 44, 45]. Perceived closeness to the decedent is associated with greater adverse psychological effects in service members and veterans [41, 42, 44].

5 | Suicide in the General Population

In the general population, risks for depression and suicidality extend back to childhood maltreatment in some people, such as sexual abuse, physical abuse, and emotional abuse [46, 47]. Suicidality is a clinical feature of major depressive disorder, and for people with depression additional risk factors for *completing* suicide include male gender, a family history of a psychiatric illness, prior suicide attempts, having more severe depression, experiencing hopelessness, having comorbid anxiety, and misusing alcohol or drugs [48]. People with sleep disturbances are more likely to experience suicidal ideation, attempts, and completed suicide [49, 50], and sleep disturbances in people who have depression are associated with greater odds of suicidality [51, 52]. There is an association between suicidality and migraine [53, 54], and having comorbid generalized anxiety and migraine is associated with increased odds of suicide ideation [55]. Suicidal ideation is more common in people who misuse drugs [56], alcohol [57], or opioids [58, 59] and drug use disorders are strong *proximate* risk factors for completed suicide [60].

In addition to depression and substance abuse, common precipitating events for suicide include physical health problems, occupational or financial problems, criminal or legal proceedings, intimate partner problems, and a life crisis [61]. Hopelessness [62, 63], problems generating adaptive solutions to life problems [64], perceived burdensomeness [65], and mental pain [66] are all high-risk patterns of thinking or cognitive states for suicidality. Suicidality in older adults is associated with functional disability and specific medical problems such as arthritis, pain, chronic obstructive pulmonary disease, liver disease, malignant diseases, and neurological disorders [67]. Pain is a stronger predictor for suicide for men than women [68]. The above risk factors, and more, are important to consider when trying to determine whether participation in contact and collision sports during youth and early adulthood is associated with later-in-life suicidality and suicide.

6 | Conclusions

Suicidality during life and suicide as a manner of death were not considered to be clinical features of CTE between 1929 and 2009 as illustrated in reviews of the literature published by Roberts in 1969 [69], McKee and colleagues in 2009 [8], and Victoroff in 2013 [70]. In the book published by Roberts in 1969, he did not describe suicidality or suicide to be a clinical feature in his review of the literature or in his large study of boxers [69]. Decades later, in their review of the neuropathology and clinical features of CTE, McKee and colleagues did not describe suicidality to be a clinical feature [8]. Victoroff also reviewed the literature, and he did not identify suicidality or suicide as a clinical feature [70].

Between 2010 and 2014, some researchers speculated that suicidality and suicide might be clinical features of CTE [8, 71–76], although there were no studies that examined and found such an association. Prior reviews of the literature relating to suicide and CTE, published between 2013 and 2020, concluded that there was not good scientific evidence to support an association [77–81]. This present review paper illustrates that there is an abundance of published evidence suggesting that suicide is not related to CTE-NC.

7 | Perspective

Concern about later-in-life brain health is a major issue for former professional football players. In a survey of former NFL players, more than one-third reported that they were 'extremely concerned' about developing CTE (39.5%) [82]. In another survey of former NFL players, one-third (34.4%) reported that they believe they have CTE [83]. Concerningly, those former NFL players who believe that they have CTE were more likely to report suicidality (25.4%) compared to former NFL players who did not believe they had CTE (5.0%). When working with current and former professional and amateur athletes who are experiencing depression and suicidality, it is important to instill hope and confidence that with treatment they can improve. If they express concern about having 'CTE,' clinicians might wish to convey the knowledge gap endorsed by the Concussion in Sport Group, that it is *not known whether CTE-NC causes specific neurological or psychiatric problems* [3]. The best available evidence suggests that CTE-NC is not associated with depression, suicidality, or suicide.

Disclosure

The author serves or has served as a scientific advisor for NanoDx (formerly BioDirection Inc.), Sway Operations LLC, and Highmark Inc. He has a clinical and consulting practice in forensic neuropsychology, including expert testimony involving individuals who have sustained mild TBIs (including athletes and including the topic of suicide). He has received research funding from several test publishing companies, including ImPACT Applications Inc., CNS Vital Signs, and Psychological

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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